



Pitches	50.6%	14.2%	23.3%	57.1%
n = 403	Swing%	Whiff%	CSW%	Zone%

KEY FINDING

Low HAA = 3× more whiffs than High HAA

Low HAA (≈1.08°) generates 25.9% Whiff% — 3× the rate of High HAA (8.5%).

When Beau’s fastball exits the hand at a lower horizontal approach

angle, it arrives from the 1B side and finishes outside to RHH.

That deceptive angle makes it extremely difficult to square up,

driving the dramatic difference in swing-and-miss across folds.

HAA and PlateLocSide share a 0.90 correlation — the approach

angle dictates where the ball ends up, not the other way around.

Low HAA (25.9% Whiff) Mid HAA (11.4% Whiff) High HAA (8.5% Whiff)

TARGET RELEASE POINT

Current RelSide: -2.35 ft

Target RelSide: -2.25 ft

Delta: Δ = +0.10 ft (≈1.2 in toward 1B/glove side)

Expected outcome: Whiff% → ~25.9% (from 14.2% overall)

For a LHP, shifting release toward the 1B / glove side

lowers HAA → ball arrives from 1B side → stays outside

vs RHH — the most effective approach angle.

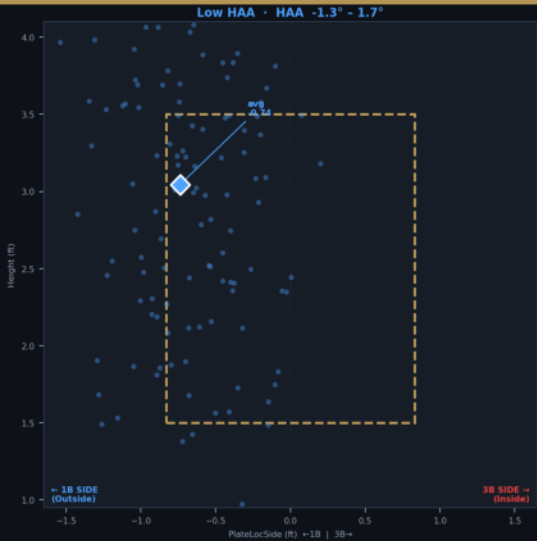
Guardrail: Zone% drops to 41.5% in Low-HAA fold.

Pair with location discipline or pitch-to-contact mix.

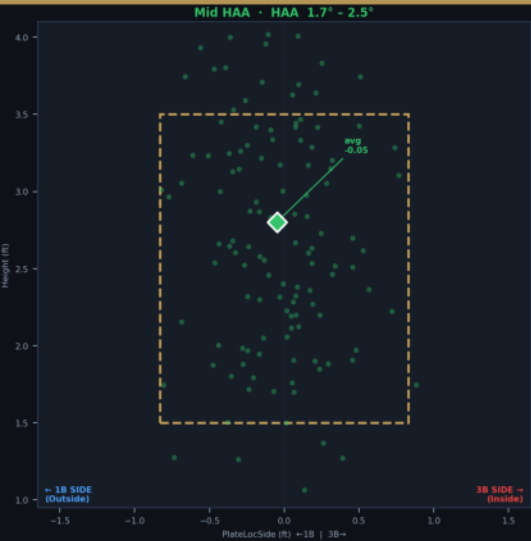


Elson, Beau (LHP) — Fastball HAA Fold Analysis · vs RHH Only

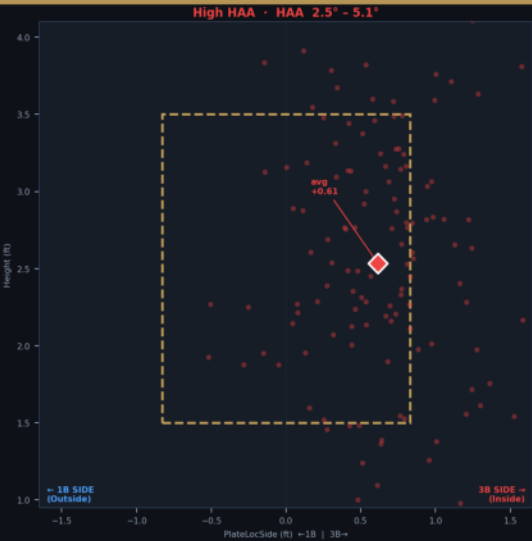
Fastballs split into 3 equal HAA terciles · Positive HAA = ball arriving from 3B side (inside to RHH) · Negative/low HAA = ball arrives from 1B side (outside to RHH) · Low HAA most effective



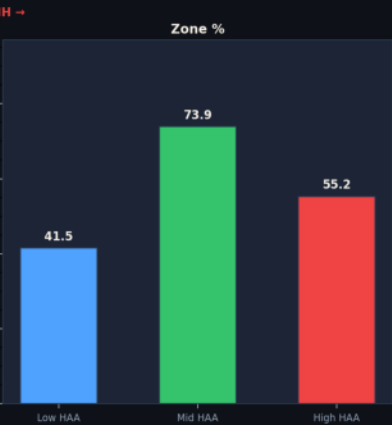
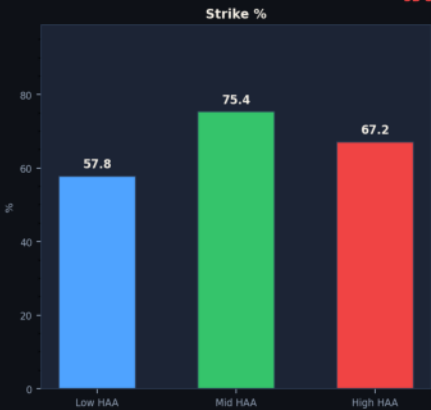
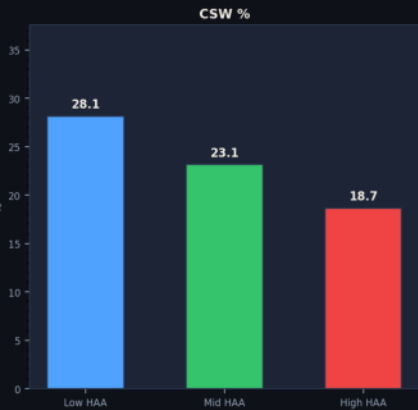
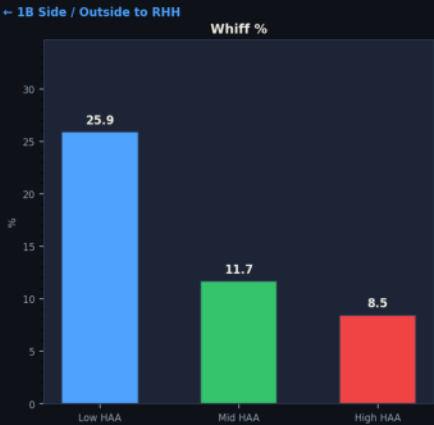
Outside (1B side)	
N (vs RHH)	135
Avg HAA	1.08°
Avg Velo	89.3 mph
Avg Rel Side	-2.25 ft
Avg Location	-0.74 ft
Strike%	57.8%
Zone%	41.5%
CSW%	28.1%
Whiff%	25.9%
Ball stays 1B-side / away from RHH → Most whiffs, highest CSW	



Middle	
N (vs RHH)	134
Avg HAA	2.12°
Avg Velo	89.2 mph
Avg Rel Side	-2.35 ft
Avg Location	-0.05 ft
Strike%	75.4%
Zone%	73.9%
CSW%	23.1%
Whiff%	11.7%
Ball works middle of plate → Moderate outcomes	



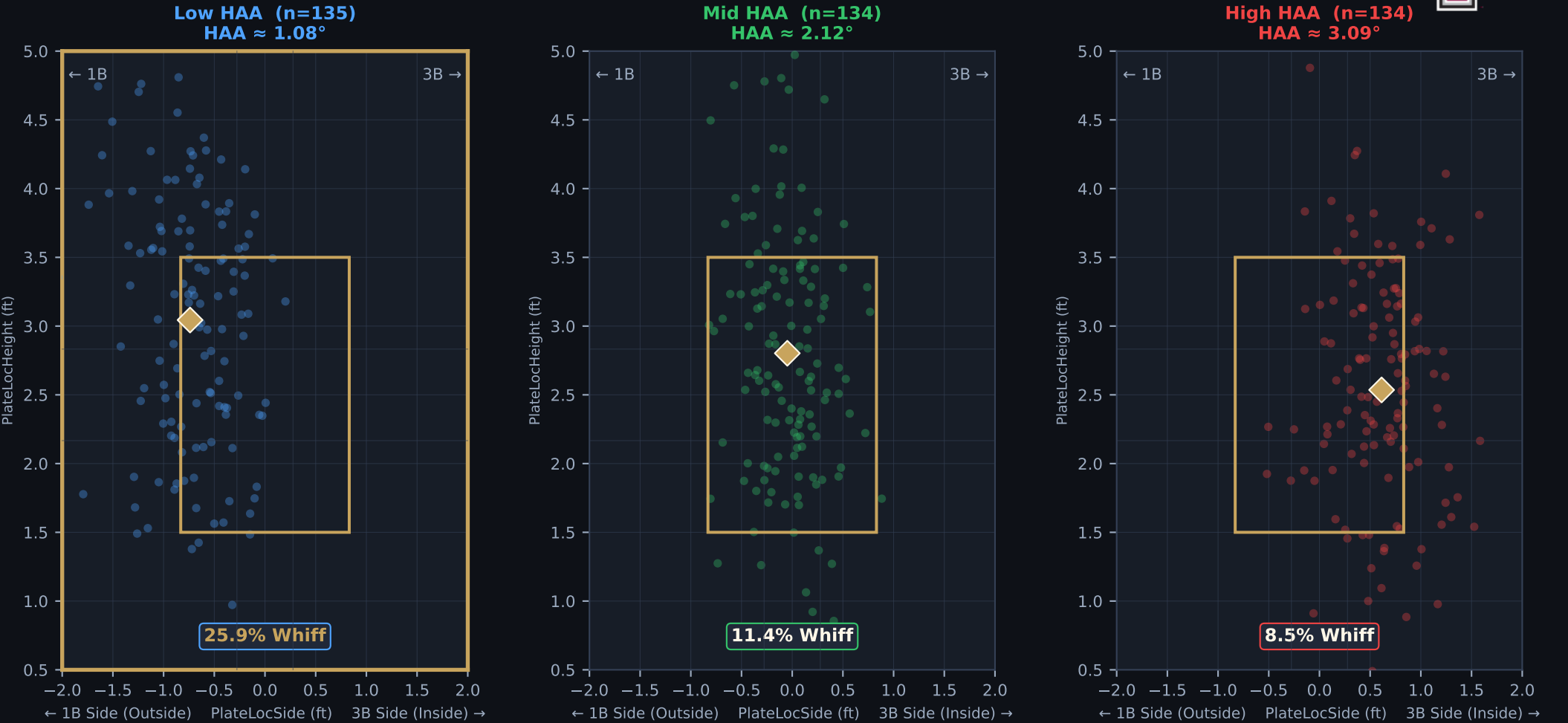
Inside (3B side)	
N (vs RHH)	134
Avg HAA	3.09°
Avg Velo	88.9 mph
Avg Rel Side	-2.44 ft
Avg Location	+0.61 ft
Strike%	67.2%
Zone%	55.2%
CSW%	18.7%
Whiff%	8.5%
Ball runs inside (3B-side) to RHH → Harder to generate whiffs	



Low HAA · Outside (1B side) · HAA -1.3° - 1.7° Mid HAA · Middle · HAA 1.7° - 2.5° High HAA · Inside (3B side) · HAA 2.5° - 5.1°

Three HAA tercile folds (Low / Mid / High) vs RHH | Low fold: HAA ≈ 1.08° → 25.9% Whiff% · Mid fold: HAA ≈ 2.12° → 11.4% Whiff% · High fold: HAA ≈ 3.09° → 8.5% Whiff%

Beau Elson · Zone Location by HAA Fold (vs RHH)



Fold Comparison — Fastball vs RHH													
Fold	N	HAA	RelS	RelH	Swing%	Whiff%	CSW%	Zone%	ZTake%	ZSwing%	OSwing%	ZCon%	OCon%
Low	135	1.08°	-2.25	5.66	40.0%	25.9%	28.1%	41.5%	35.7%	64.3%	22.8%	83.3%	55.6%
Mid	134	2.12°	-2.35	5.61	59.0%	11.4%	23.1%	73.9%	26.3%	73.7%	17.1%	90.4%	66.7%
High	134	3.09°	-2.44	5.58	53.0%	8.5%	18.7%	56.0%	24.0%	76.0%	23.7%	94.7%	78.6%



RUBBER / RELEASE RECOMMENDATION

Current RelSide: -2.35 ft

Target RelSide: -2.25 ft

Required Shift: Δ = +0.10 ft ≈ 1.2 in toward 1B/glove side

Expected Outcome: Whiff% → ~25.9% (from 14.2% overall)

For a LHP, shifting release toward the 1B / glove side

lowers HAA → ball arrives from 1B side → stays outside

vs RHH — the most effective approach angle.

9-Zone Whiff% (All vs-RHH Fastballs)

High	33.3%		30.3%		12.0%	
Mid	11.8%		3.6%		0.0%	
Low	8.3%		6.2%		8.3%	
		1B Side (Outside)			Middle	3B Side (Inside)

STRATEGIC READ PER FOLD

LOW HAA (≈1.08°)

- Ball arrives from 1B side, finishing outside to RHH.
- 25.9% Whiff% — 3× the rate of High-HAA zone.
- Zone% is 41.5%; effective even when missing slightly off plate.
- Primary use: put-away pitch / two-strike count; must spot carefully.

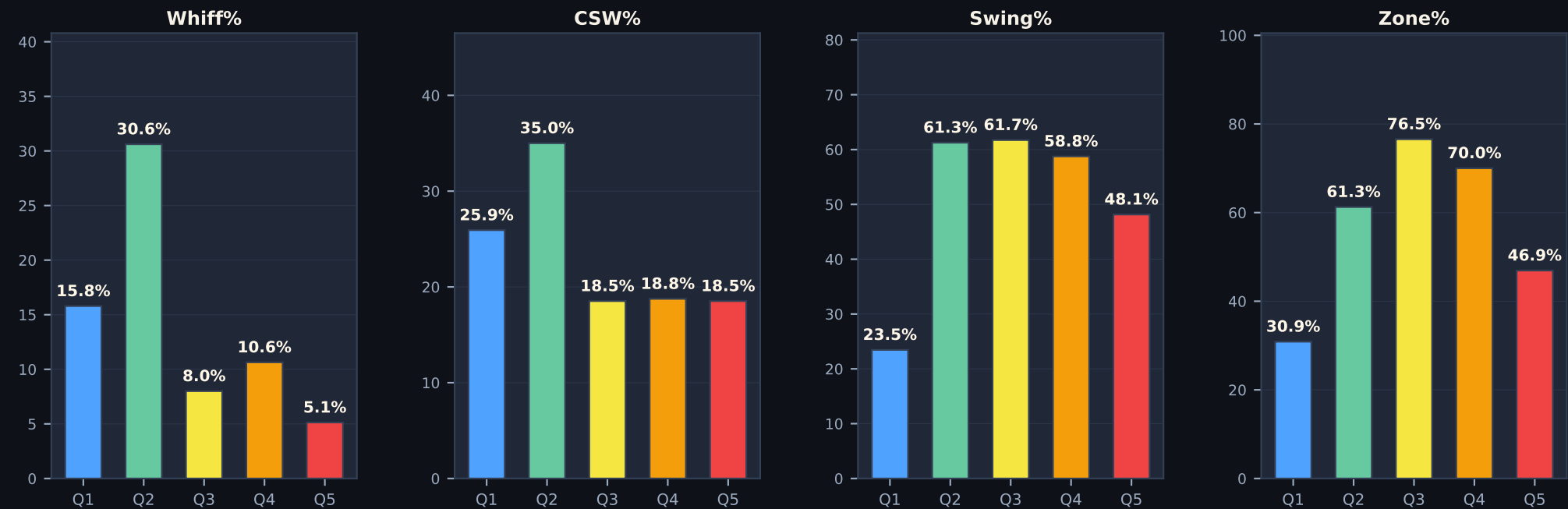
MID HAA (≈2.12°) — NEUTRAL

- Ball centers in zone (PlateLocSide ≈ -0.05 ft). High Zone% (73.9%).
- Moderate swing-and-miss (11.4%). Good for early counts / strike-one.
- Use to establish zone presence before going to Low-HAA for whiff.

HIGH HAA (≈3.09°) — AVOID FOR WHIFFS

- Ball runs inside to RHH (PlateLocSide +0.61 ft). Harder to miss.
- Only 8.5% Whiff% — hitters can track and barrel the pitch.
- Zone% (56%) moderate. Risk of hard contact on middle-in fastballs.
- Use sparingly as a complementary location for pitch-mix variety.

Beau Elson · 5-Fold HAA Analysis (Quintiles vs RHH)



5-Fold (Quintile) Comparison — Fastball vs RHH													
Fold	N	HAA	RelS	RelH	Swing%	Whiff%	CSW%	Zone%	ZTake%	ZSwing%	OSwing%	ZCon%	OCon%
Q1 (Lowest)	81	0.78°	-2.25	5.68	23.5%	15.8%	25.9%	30.9%	52.0%	48.0%	12.5%	91.7%	71.4%
Q2	80	1.62°	-2.28	5.63	61.3%	30.6%	35.0%	61.3%	30.6%	69.4%	48.4%	79.4%	46.7%
Q3	81	2.12°	-2.35	5.61	61.7%	8.0%	18.5%	76.5%	21.0%	79.0%	5.3%	91.8%	100.0%
Q4	80	2.58°	-2.38	5.60	58.8%	10.6%	18.8%	70.0%	23.2%	76.8%	16.7%	90.7%	75.0%
Q5 (Highest)	81	3.37°	-2.47	5.57	48.1%	5.1%	18.5%	46.9%	26.3%	73.7%	25.6%	100.0%	81.8%

Q1 (lowest HAA, avg 0.78°) → outside to RHH Q5 (highest HAA, avg 3.37°) → inside to RHH